**7** Which equation of uniformly accelerated motion can be derived using only the gradient of a velocity–time graph?

**A** $s = \frac{1}{2}(u + v)t$

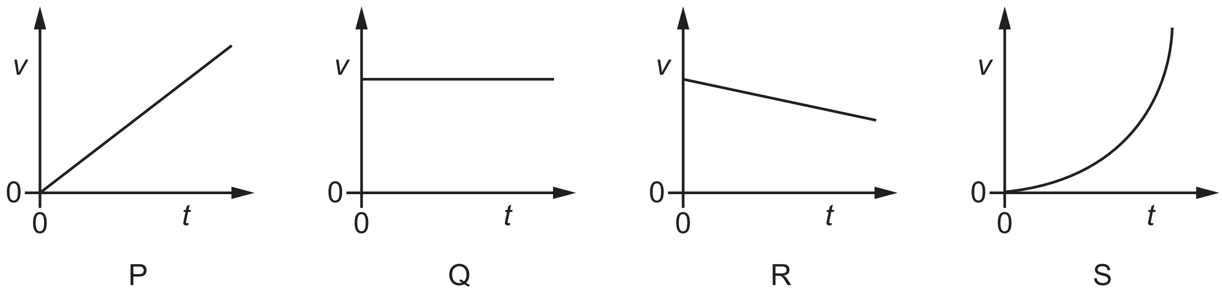
**B** $s = ut + \frac{1}{2}at^2$

**C** $v = u + at$

**D** $v^2 = u^2 + 2as$

**8** An object is projected horizontally from a table at time $t$ = 0. The object falls in a uniform gravitational field. Air resistance is negligible.

Graphs P, Q, R and S are velocity–time graphs.




Which graphs represent the horizontal and vertical components of the velocity of the object?

| | horizontal | vertical |
|---|---|---|
| **A** | Q | P |
| **B** | Q | S |
| **C** | R | P |
| **D** | R | S |

**9** What is always conserved in elastic collisions?

| | total kinetic energy | total velocity |
|---|---|---|
| **A** | yes | yes |
| **B** | yes | no |
| **C** | no | yes |
| **D** | no | no |